A Cognitive Model and Cognitive Rationale for Natively Multilingual Robots

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Abstract

Although the most obvious way to make cognitive robots multilingual would be to use machine translation (MT), that is not the best option due to both the nature of language and the nature of the people who use it. A better option is to make cognitive robots *natively multilingual* in each language of interest — an approach that is straightforward for the cognitive robots called LEIAs. This paper explains why MT is an insufficient solution, and it describes the cognitive model that supports LEIA-style native multilinguality. Proof of concept is provided via simulation in the domain of shipboard mechanics.

1 Introduction

Over the past half century, the standard in natural language processing has been to push empirical methods to their limits and then either postpone the problems they cannot solve or attempt to deal with them using knowledge-based patches. This mindset has recently extended to large language models, despite their recalcitrant hallucinations. If one were to apply this approach to developing multilingual cognitive robots, it would likely involve preparing them to communicate in one language and then using an off-the-shelf machine translation (MT)[1] engine for other languages, possibly supplemented with fix-up rules.[2]

A better solution, both scientifically and technologically, is to create native language processing capabilities for each language a cognitive robot will speak. In this way, the robot will be working within the same holistic language system as its human partner. This makes all the difference in making cognitive robots not only effective communicators, but also trustworthy and reliable partners.

[1] All current MT uses empirical methods, be they grounded in an LLM or not. For a survey, see Ataman et al. (2025).

[2] Using language models to replace cognitive modeling entirely will not work for cognitive robots, as demonstrated in Oruganti et al. (2026). Similarly, cognitive models trained on English cannot be ported to other languages without introducing significant errors, as discussed in Section 5.

This approach to multilinguality is being developed for the cognitive robots called LEIAs[3] (McShane & Nirenburg, 2021; McShane, Nirenburg, & English, 2024). Here, we describe theoretical and methodological principles of multilingual cognitive modeling that are being implemented in a particular way in LEIAs, but can also be applied to any cognitively grounded agent system.

This paper argues in favor of *native* multilinguality over reliance on MT due to the nature of languages, the nature of people, the limitations of MT, and the opportunities of holistic cognitive modeling. Fig. 1 presents key issues, which we will briefly explain as a warm-up.

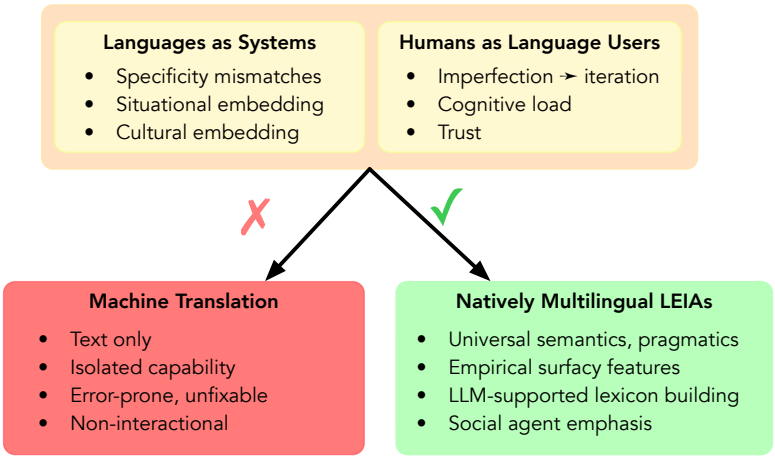


Fig.1 Key modeling issues related to multilingual robots.

Languages as Systems. Natural languages are systems that are nested in larger systems of multimodal communication and culturally embedded action. They cannot be reduced down to the words on page, and, even if they could, the words on a page do not neatly align across languages.

Humans as Language Users. Spoken language is not textbook perfect, but the language system accounts for that — as by offering recovery strategies when a message isn't fully understood. For example, people can wait and see if things become clear in time, they can ask a clarification question, or they can ignore the information if they consider it non-essential. People can't be expected to change their communication-oriented behavior — or any behavior, for that matter — based on the limitations of technology. When intelligent systems aren't up to par, people lose both trust and their willingness to engage.

Machine translation (MT). MT treats language as text, which can sometimes work in written genres but falls apart for spoken language in real-world contexts. By isolating text from non-linguistic perception, reasoning, and action, and by pushing out translations in a unidirectional, non-interactive manner, MT systems make errors that will not be solved by additional compute.

[3] The acronym originally stood for Language-Endowed Intelligent Agents but has recently been recast as Language, Embodiment, Intelligence, Agency in order to reflect the expansion of our lab's program of R&D to cognitive robotics. See https://leia-lab.com/ for details and video demonstrations.

Natively multilingual LEIAs. Agents can be prepared to communicate like humans if they are modeled to emulate them. This requires symbolic cognitive modeling that integrates language processing into overall cognition and action. When such modeling is approached in the style of LEIAs, it has the added benefit of organically enabling native multilinguality. That is, the human-inspired microtheories of language understanding and generation that were developed to support English-language communication by LEIAs naturally accommodate additional languages — which is not surprising seeing as the theory of Ontological Semantics that underlies LEIA operation was originally developed to support interlingual MT (Nirenburg & Raskin, 2004; Nirenburg, Somers, & Wilks, 2003).

1.1 Brief Background on LEIAs

LEIAs are neurosymbolic, multimodal cognitive-robotic systems that are implemented in the HARMONIC cognitive architecture (Oruganti et al. 2024). HARMONIC integrates cognitive (strategic) and robotic (tactical) layers. The cognitive layer primarily relies on knowledge-based modeling to support reliability and transparency. The robotic layer primarily relies on machine learning, which is effective and sufficient since the associated capabilities — such as moving a robotic arm or avoiding collisions — need not be computationally transparent or explainable. The components of the cognitive and robotic layers function both independently and interactively.

LEIAs represent an approach to Agentic AI that we call OntoAgentic AI. Whereas typical Agentic AI systems use language models both as the orchestrator and for support functions, OntoAgentic AI uses a LEIA as the orchestrator and leverages both LEIA agents and language-model-based systems for support functions. OntoAgentic AI, therefore, offers reliable, explainable control of overall system operation as well as the cognitive operation of each individual LEIA.

LEIA cognition orients around meaning, which is defined in terms of an unambiguous, language-independent ontology, following the theory of Ontological Semantics. When LEIAs interpret stimuli, reason, or plan, they do it in terms of ontologically-grounded meaning representations.[4] For example, whether an agent sees that the valve on a particular pipe is off or is told that that valve is off, it will store the following to episodic memory:

VALVE-12
  PART-OF-OBJECT PIPE-43
  ON-OFF-STATUS off
  TIME 2026-04-02T12:35

Depending on how the agent perceived this information, the meaning representation will be decorated with different metadata. For example, if someone said, “The valve is off”, then the frame above will be the THEME of an instance of DECLARATIVE-SPEECH-ACT; the AGENT (speaker) and BENEFICIARY (hearer) will be indicated; and the lexical senses that were used to generate the analysis will be indicated. If the agent wanted to convey the idea “The valve is off”, it would start the

[4] See McShane and Nirenburg (2021, section 2.8.1) for reasons why using an ontological metalanguage is a better option than using natural language as proxy for meaning.

language generation process from a meaning representation like this, which is the output of content specification.

That should suffice as general background. More about language modeling in LEIAs will be explained in Section 3, but first let us look at why the nature of human languages resists the text-centric constraints of empirical MT.

2 Language Resists MT by Its Very Nature

Languages are systems that are embedded in larger systems of culturally-sensitive, multimodal communication. During normal interactions, people perceive multimodal stimuli, they build a shared situation model, they reason about the task and each other, they continually monitor how well they understand each other, and, when needed, they seek and provide clarification. Off-the-shelf MT doesn't have access to any of this, making it error-prone in embodied contexts, despite the fact that it often performs quite well in text-based genres. Here we see the same quality dichotomy as found with large language models overall: when they work well, they can be very impressive, but hallucinations remain a serious and recalcitrant risk.

We will explain why natural languages resist MT, particularly for embodied agents, using three categories of examples: specificity mismatches, the situational embedding of language, and the cultural embedding of language.

Specificity mismatches. Languages differ markedly with respect to which linguistic features they do and do not express. In other words, they are highly decorated in some ways and bare-bones in others. This means that translating in one direction often loses features while translating in the other direction requires inferencing about them, both of which can lead to failures of communication. For example:

- Some languages have **gendered inanimate objects** whereas others don't. The French *Donne-moi la$_{FEM}$ bleue$_{FEM}$* 'Give me the blue one' makes it clear that the object must be of feminine gender because *la bleue* is grammatically feminine. So, if a feminine 'cup' (*tasse$_{FEM}$)* and a masculine 'glass' *(verre$_{MASC}$)* are in view, this expression can only refer to the cup. However, if this sentence is translated into English, the gender-based information is lost, potentially introducing referential ambiguity. On the flip side, translating English 'the blue one' into French requires selecting a gender, which requires understanding which referent the speaker had in mind.
- Some languages have **virility distinctions**. The Polish *One$_{NON\text{-}VIRILE}$ trzebują twojej pomocy* 'They$_{NON\text{-}VIRILE}$ need your help' can only refer to a group without men. So, if there is exactly one all-women group in the context, this is an unambiguous utterance. However, translating the Polish pronoun *one* into English 'they' loses this disambiguating information. On the flip side, translating from English into Polish requires knowing who the speaker has in mind.
- Some languages indicate **clusivity**, which means that they distinguish between inclusive and exclusive *we*. In Telugu *manam* refers to a group that includes the speaker, the addressee and possibly others, whereas *mēmu* refers to the speaker and others, but necessarily *not* the addressee. Translating into English *we* erases this distinction, whereas translating *we* from English into Telugu requires inferring it.

- Some languages permit **subject ellipsis**, which means that grammatical subjects can be left out when they are understood from the context. The Polish *Wchodzi*$_{3.SG.PRES.}$ 'Is-coming-in' doesn't indicate whether a male or a female person is coming in. Translating into English requires choosing between *he* and *she*, whereas translating from English into Polish loses that disambiguating pronoun.[5]

These examples all happen to involve referring expressions, which are quick to explain and relevant for robots. However, the comparative linguistic literature is chock full of other features that play out differently across languages—tense, aspect, evidentiality, counterfactuality, spatial deixis, expression of motion events, politeness (see below), and many others.

One question that might come to mind is how people deal with instances of underspecification *within* a language system? The short answer is that they know how to work around them, as by using expressions that are not ambiguous. However, if ambiguities arise, then it natural for the interlocutor to ask for clarification. For a more detailed answer, we need to consider language within its situational embedding.

Situational Embedding. When speakers use expressions that might be understood variously — such as English *they, we,* or *the blue one*— they are making a judgment call that the context makes the referent clear to the interlocutor. But there can be various eventualities:

- If the speaker realizes after speaking that the referent might not be clear, then they can clarify: for example, "*We're taking Friday off — that is, Dan and I."*
- If the hearer doesn't understand an underspecified referent, they can ask about it: "*We're taking Friday off" "Who is?"*
- If the hearer thinks they've understood but it becomes clear that they haven't, then various repair strategies are available: *"We're taking Friday off." "Great! Tell Jim I hope you guys have a good time." "Actually, it'll be Dan and I. But I'll pass him the good wishes!"*

All of these options are baked into language systems, and when children acquire a language, they learn the whole system. To put this in the context of Fig. 1, human communication is imperfect but iterative—an opportunity that is lost in an MT-based approach to multilinguality for robots.

What would not be acceptable to most people would be for robots to act strangely or outright make mistakes based on MT errors. For example, if a speaker of Telugu tells a robot "We$_{\text{EXCLUSIVE}}$ are going to take a 10 minute break", this clearly excludes the robot. If the robot's MT system translates this into English as just 'we', then the robot won't know whether it should or shouldn't take a break. If it then asks. 'Does that include me?', the question will sound bizarre — after all, the Telugu speaker made that clear. If, by contrast, the robot *doesn't* ask for clarification but guesses that the message includes itself, that will be another sort of mistake and it will lose possibly valuable minutes of work time. Either way, the robot's wrong behavior will have two negative impacts. First, it will increase the cognitive load on the humans, turning language use into a problem rather than

[5] In Polish, the pronoun can be overt as well, but only in contrastive and other marked contexts; the default is to elide it. For discussion of subject ellipsis in Russian and Polish, see McShane (2009).

an effortless part of a day's work. Second, it will cause the human to lose trust since, if a robot can't even understand what is plainly said to it, why should it be expected to do anything intelligent?

Cultural embedding. When speakers of English think about the interaction of language and culture, one example that likely comes to mind involves politeness and honorifics, for which Japanese, Korean, and Hindi have famously nuanced systems; and even many European languages have two-way distinctions for *you* (formal/informal). Translating *from* these languages into English loses nuance, whereas translating *into* these languages requires careful analysis and risks a selection error — which, in many cultures, will not be viewed as a minor slip. In fact, speakers of such languages can feel uncomfortable speaking English because using *you* with everyone feels rude. In order to get these distinctions right, a system needs a cognitive model of social relations in the language/culture as well as episodic memory that includes relevant information about all interlocutors.

Another cultural norm that plays out very differently across languages involves mitigated speech, which is softened, less direct language. Mitigated speech is used in order to avoid imposition and conflict, but there is a trade-off: it often decreases the clarity of the message. For example, saying "Get me a cup of coffee" is direct, and might be interpreted as impolite, but it is clear; by contrast, "I'd love a cup of coffee" leaves it unclear whether the interlocutor is expected to do something about that. Within a particular language/culture pairing, people understand norms of usage as well as strategies of recovering from excessive ambiguity. Continuing with the second example, the interlocutor might ask, "Would you like me to get you one?"

However, across languages and cultures there can be ruptures that can have disastrous consequences. A famous example involves Avianca Flight 52 from Bogotá to New York in 1990 (Ogiermann, 2009; Tajima, 2004). The crew, native speakers of Spanish, knew they were disastrously low on fuel but used mitigated speech, "running low on fuel," rather than the word "emergency" when communicating with the New York air traffic controllers. The flight was not cleared to land for another 39 minutes and crashed, killing 73 people. Within the cultural norms of American English, the air traffic controllers expected a more direct formulation of urgency and a more direct demand for help than what was conveyed by the native speakers of Spanish.

To recap this section, each language is a complex system embedded in larger systems of communication and culture. The surfacy aspect of language — the words on a page — cannot be extracted from the whole, translated as if they were an isolated artifact, and then plopped into another language's communication and cultural system: it just won't work. The fact that MT systems exist and can perform pretty well for some language pairs in formal written genres and carefully selected examples of speech does not mean that MT is a long-term solution for cognitive robots.

3 Modeling Native Multilinguality

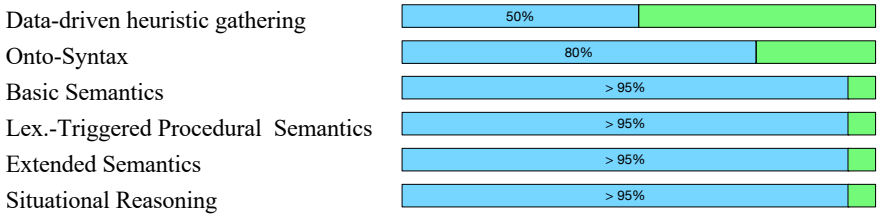
Central to modeling multlinguality in cognitive robots is distinguishing between language-specific and language-independent functionalities. **Language-specific** functionalities cover surface features of language, such as segmentation, morphology, and syntax. **Language-independent** functionalities involve semantics, pragmatics, and real-world reasoning. LEIAs already provide language-independent functionalities that can be tweaked as needed for individual language/culture combinations. As for language-specific functionalities, they can be handled pretty well by empirical tool-

sets, which is not surprising considering that (a) the majority of work on natural language processing for the past 70 years has focused almost exclusively on these, and (b) language models are good at capturing surface regularities over vast corpora. This integration of empirical and symbolic processing maximizes the use of modern technology while acknowledging its limits and upholding the reliability of agents. We will now look at how language understanding and generation play out in multilingual LEIAs.

3.1 Modeling Language Understanding

Language understanding is modeled as the six stages shown in Table 1 (McShane & Nirenburg, 2021; McShane, Nirenburg, & English, 2024).

Table 1. A rough approximation of the language-independent (blue) versus language-specific (green) knowledge and processing for each stage of language **understanding**.



Broadly speaking, the first two stages—which involve the surfacy features of language—reflect a mix of language-specific and language-independent processing, whereas the latter four stages—which cover semantics and pragmatics—are almost completely language independent.

The first stage is **Data-driven Heuristic Gathering**, which uses data-driven toolsets specific to each language to compute sentence boundaries, lemmas, part-of-speech tags, morphological features, a constituency parse, a dependency parse, named entity information, and some aspects of textual coreference. Currently we use the Berkeley Neural Parser (benepar), which offers support for many languages (Kitaev, Cao, & Klein, 2019).[6] The output of benepar is imported into LEIA operation using a custom wrapper. Although the actual tools are different for every language, they are created by the same machine-learning based methods, which explains our 50-50 estimation of language independence. That is, the methods are language independent but the resulting tools are language specific. So far we have experimented with benepar's tools for French and Polish because they provide the same level of support as for English. For other languages, some functionalities, such as tokenization, need to be sourced separately and imported in a customized fashion. This is not hard, it simply requires engineering effort that is not currently funded.

During **Onto-Syntax** the LEIA creates a bridge between the syntactic parse tree and the LEIA's semantic lexicon. Basically, it says "If I only look at syntax for the moment, here is how word senses in the lexicon might be combined to cover the input." This stage involves syntactic transformations and combinatorics. The vast majority of operations are language independent since the workhorse of Onto-Syntax is a program that combines lexical senses in syntactically legal ways.

[6] Examples of other data-driven toolsets that can provide pre-semantic heuristics are Kondratyuk & Straka (2019) and the Stanford Stanza toolset (https://stanfordnlp.github.io/stanza/).

However, some language-dependent work does need to be done to support multilingual processing: a lexicon for each language needs to be acquired and the inventory of transformations needs to be tweaked, as by adding transformations to account for subject ellipsis in languages that permit it.

Acquiring a LEIA-style semantic lexicon is not as labor-intensive as it might sound because lexicons for other languages can be bootstrapped from the English lexicon semi-automatically. Two things facilitate this process. First, most of the information in the lexicon is actually language-independent, as explained in McShane, Nirenburg, and Beale (2005) and illustrated by the example in Table 1. When porting this English verb sense to Polish most of the information stays the same except for the following: a) different Polish senses needed to be created for the perfective and imperfective aspects; b) the aspect of each needs to be recorded as metadata; c) the meaning of the aspect needs to be recorded as a value of PHASE; d) the particle *się* needs to be added to the syntactic structures; and e) translations of the head words need to be provided (*uczyć, nauczyć*). The human-oriented metadata (definition and example) can be optionally edited: it is not required for processing.[7] These kinds of modifications are entirely standard for English-to-Polish bootstrapping and language models are well suited to speeding along the process, leaving people to check rather than manually acquire senses.[8]

Table 1. Bootstrapping a Polish word sense from an English one.

learn-v1	**uczyć-v1**	**nauczyć-v1**
def. to acquire a skill ex. Paul learned to ski.	def. rozwijać umiejętność ex. Paul uczy się jeździć na nartach. **aspect impfv**	def. rozwinąć umiejętność ex. Paul nauczył się jeździć na nartach. **aspect pfv**
syn-struc subject $var1 v $var0 xcomp $var2	syn-struc subject $var1 v $var0 **particle $var3 (root się)** xcomp $var2	syn-struc subject $var1 v $var0 **particle $var3 (root się)** xcomp $var2
sem-struc ACQUIRE-SKILL AGENT ^$var1 THEME ^$var2 ^$var2 AGENT ^$var1	sem-struc ACQUIRE-SKILL AGENT ^$var1 THEME ^$var2 PHASE IMPFV ^$var2 AGENT ^$var1 **^$var3 (null-sem+)**	sem-struc ACQUIRE-SKILL AGENT ^$var1 THEME ^$var2 PHASE PFV ^$var2 AGENT ^$var1 **^$var3 (null-sem+)**

The time savings of porting LEIA lexicons across languages rather than building new lexicons from scratch becomes ever more impressive when the meaning to be described is inherently tricky. For example, the meanings of some words — such as abstract notions like *privacy* — are difficult to describe, and such problems need to be solved when acquiring the first lexicon. Similarly, some words must be analyzed procedurally — such as applying the meaning of *very* to the value of a scalar attribute — and that program must be written when building the first lexicon.

[7] In order to work quickly and efficiently with LEIA resources, developers need to know English because ontological concept names are based on English, even though concepts and words are entirely different things. That is, a LEIA would operate identically if the concept ADMIRE were named $ja7, but the people working with the system would not understand the corresponding meaning representations at the center of LEIA cognition.

[8] For our experimentation with using LLMs as knowledge workers, see Nirenburg et al. (2026).

Of course, we are not saying that all of the constructions in any lexicon will be directly portable to, or necessary in, other languages; and, clearly, typological similarity is a plus. What we *are* saying is that (a) a whole lot of lexical content is reusable across languages, and (b) our construction-based orientation means that constructions that might make a language seem much different than English — e.g., impersonal constructions in Slavic languages — are operationally not different at all. So, a lexicon of the type we are describing erases a lot of the surface differences found across languages.

As regards transformations, they account for such phenomena as passivization, verbal complements, ellipsis, question formation, and much more. Many transformations, such as passivization, work much the same across languages, with only minor lexical tweaks. Other transformations, such as argument ellipsis in Slavic, can be readily handled as variations on the theme of nominal coreference. Again, surface-level differences across languages are smoothly handled using procedures that recognize their linguistic similarity.

OntoSyntax outputs syntactically legal combinations of lexical senses and passes them to the stage called **Basic Semantics**, which focuses on lexical disambiguation and the establishment of the semantic dependency structure for individual sentences. This process uses not only the information from lexical senses but also the property-based descriptions of concepts in the ontology. Basic Semantics operates exclusively over concepts and it involves a single, generic processing algorithm, making it completely (as far as we can tell) language independent. Table 1 leaves 5% as being potentially language dependent just in case we have not foreseen something. Basic Semantics outputs one or more candidate meaning representations.

The next stage, **Lexically-Triggered Procedural Semantics,** takes care of some of the special cases that were not addressed by Basic Semantics. It runs procedural semantic routines that were listed in the lexical senses used by an analysis. Such procedures cover many different kinds of phenomena, from textual coreference (*she*) to scalar modification (*very* hot) to words whose meanings guide the interpretation of a whole utterance (*respectively*). An important modeling decision is that these functions are not specific to the words of any particular language. Instead, they are recorded as procedural attachments to ontological concepts, so any language that has a word or phrase that requires the given kind of reasoning can point to the same function in the ontology — in some cases, with language-specific parameterization. An example where no language-specific input parameters are needed is RESOLVE-SCALAR-VERY, which figures out, for example, that *very cold* takes the meaning of *cold* "TEMPERATURE .2" and moves it down the scale by .1 to "TEMPERATURE .1". This works exactly the same for any language. An example in which language-specific input parameters *are* invoked is COREF-SUBJ-PRO, which seeks a textual coreferent for an overt or elided subject pronoun. Although many of the heuristics used in this process are stable across languages, there can be language-specific modifications. As mentioned above, Slavic languages permit subject ellipsis, whose usage rules—which differ across Slavic languages—will not be leveraged in English.

The fifth stage, **Extended Semantics**, attempts to resolve residual ambiguities, incongruities, and underspecifications in the candidate meaning representations generated so far. It uses additional knowledge bases and algorithms, as well as a multi-sentence window of discourse. It handles such things as interpreting as-yet ambiguous nominal compounds, understanding metonymies, disambiguating objects using ontology-based reasoning, interpreting fragmentary utterances in dialogs,

and much more. Almost all of the knowledge and reasoning in this module is language independent, although some phenomena might occur more frequently and/or pose more challenges in some languages than in others: for example, nominal compounds are used more widely, and often contain more component nouns, in German and Swedish than in English.

The sixth and final stage of language understanding, **Situational Reasoning,** goes far beyond language *per se*. It leverages all of the knowledge and information available to an agent within the context of a particular, possibly multimodal, communication. Agents leverage the situation model, plan-and-goal-oriented reasoning, the interpretation of visual input, mindreading of the interlocutor, learning, etc., in an attempt to arrive at a single, precise, contextually appropriate, high-confidence analysis of a language input. Component processes include, non-exhaustively, finalizing speech act interpretation, establishing discourse relations, incorporating tricky fragments into the discourse model, grounding overt and elided referring expressions, inferring unexpressed time, and finalizing the learning of new words and phrases. Importantly, agents also assess the *actionability* of their interpretations—i.e., whether they have understood enough, with high enough confidence, to act (McShane et al., 2025a). If not, they decide what to do next, which might involve, for example, asking a follow-up question or waiting to see what happens next. The communication in the demonstration system to be described in Section 4 relies heavily on reasoning in this module.

3.2 Can Empirical Methods Handle Surfacy Processing for Every Language?

Let us loop back to a core dichotomy in our model of language understanding: Empirical tools are used to compute surfacy features, and their results feed into the symbolic processing that takes care of the rest. Using empirical tools in this way is practical on two counts. First, over the past half century, tremendous resources have been devoted to these functionalities, and the expense of reinventing all of it using symbolic cognitive modeling cannot be justified. Second, machine learning is well suited to this task — at least for languages that are typologically not too dissimilar from English, such as those native to Europe.

But what about languages that are not typologically similar to English? Will this empirical/symbolic split still work? In some cases, not as well. This is not a reason to abandon the LEIA workflow, which is both psychologically plausible and serves many languages. However, it is a reason to think about (a) how the workflow might be modified to accommodate typologically distant languages, and (b) whether such modification is merely a practical necessity or whether there is anything theoretically interesting to be learned.

A good example of a challenge to our workflow is languages that have complicated tokenization and segmentation. For example, Thai does not indicate word boundaries, and Turkish and Telugu have agglutinating morphology. In both cases, semantic analysis is needed to make the correct chunking decisions; but in the LEIA workflow, semantic information only becomes available later in the workflow.

At first glance, this might look like a show-stopping problem. But is it? Actually, no — or at least not in principle. The reason is that each stage of processing can output any number of candidate analyses. All of the candidates from stage N are evaluated during stage N+1, and any that are deemed unsuitable are discarded. By the end of stage 6, ideally, only one candidate is left. As regards tokenization and segmentation in particular, the question is whether it would be possible for empirical systems to output *all reasonable* chunking options, rather than commit to a single one that might be wrong. Current technology is not lacking in compute, so it becomes an empirical

question: How many competing options would actually need to be generated and is this too many on computational grounds? In assessing this, one must keep in mind that LEIAs are intended for practical robotic applications, not formal texts chock full of compound complex sentences.

It is noteworthy that we have already solved problems of this general class for several phenomena in English using a process we call *reambiguation*. Reambiguation overrides syntactic parsing decisions that actually require semantic heuristics. Examples include (non-exhaustively) the internal bracketing of nominal compounds that contain three or more nouns (*alligator shoe fasteners*) and the attachment of post-verbal prepositional phrases (*I saw the man with the binoculars*). Our solution is for the LEIA, during OntoSyntax, to analyze the parse returned by the empirical toolset, detect the potential for alternative analyses, and add those options back into the candidate space.

If reambiguation as currently implemented is not sufficient to handle tokenization and segmentation challenges in languages like Thai and Telugu, another possibility would be to introduce a new "Stage 0" to the LEIA pipeline, which would use the situation model to identify islands of confidence involving word boundaries, morphemes, and collocations. The empirical tokenizer could then, presumably, be configured to work around these fixed decisions.

A reasonable question at this point would be: Why does your cognitive model of language understanding not invoke the situation model earlier to begin with? In other words, why aren't LEIAs, like humans, anticipating what their interlocutor might say and using that to guide language interpretation?

The reason is more practical than theoretical. Sometimes anticipation works and sometimes it doesn't: after all, people switch topics at the blink of an eye, they say unexpected things, and the function of communication in the general case is not to confirm what someone is expecting to hear but, instead, to convey information that is not known. Moreover, building anticipation into the model would then require backtracking when expectations were wrong, which is more complex than our current approach of placing all reasonable options into the search space and culling them throughout the stages.

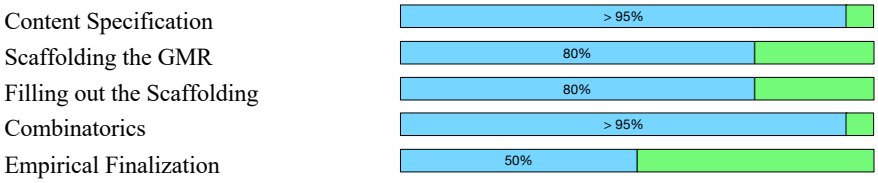
Stated differently, one might claim that humans use all of their knowledge about everything at every moment while they incrementally interpret all perceptive inputs (language, vision, haptics) simultaneously; but even if that were true, it is not a model. The challenge of cognitive modeling is to tame the chaos and make things work, which requires acknowledging that a computer does not function exactly like a human brain, and that we will never understand the latter with the sort of precision needed to accurately replicate it in silica.

To summarize, we do not claim that our model for language processing is universally applicable without modification, but that is no reason to reject it. What we have here is the well-known tension between universality, on the one hand, and utility or explanatory power, on the other (for discussion see, e.g., Evans & Levinson, 2009). For AI-oriented language processing, the good must not be jettisoned in pursuit of the perfect.

3.3 Language Generation

Language generation is modeled as the five stages shown in Table 2.[9]

Table 2. A rough approximation of language-independent (blue) vs. language-specific (green) knowledge and processing for each of the stages of language **generation**.



We will describe generation more briefly than understanding since a sister paper about it will be published in this conference's proceedings (McShane, English, & Nirenburg, 2026).

During **Content Specification**, the LEIA decides that it wants to communicate some message and it specifies the message in the ontological metalanguage as a *generation meaning representation* (GMR). This is almost wholly language independent apart from things like how to combine speech and gestures based on cultural norms.

Scaffolding the GMR involves selecting candidate lexical senses for the heads of dependency structures, working down from the root of the GMR. This is considered 20% language dependent because, while it requires a lexicon for the given language, all of the processing is generic.

Filling out the Scaffolding covers many different processes, from identifying candidate lexical senses for outstanding concepts in the GMR to handling referring expressions. It, too, is about 20% language specific due to the need for a lexicon and reference-oriented parameterization.

Combinatorics is an entirely generic computational process, for which we leave a 5% buffer for unforeseen language-specific modifications. Combinatorics outputs near-sentences, which are ordered word choices with morphological features that need to be rendered.

Empirical Finalization uses empirically-grounded toolsets to generate the needed morphological forms and select the best of the candidate sentences for the given context. Our 50/50 estimate has the same rationale as for the empirical tools used for language understanding: the methods employed are entirely generic, whereas the resulting tools are language-specific. The beauty of using an empirical tool for these purposes is that all of the candidates passed to it are guaranteed to be semantically correct, so it cannot introduce an actual error. In the worst case, it will select an option that is pragmatically suboptimal. However, testing so far suggests that its performance is actually quite good.

4 Proof of Concept Through Simulation

We tested our model in a simulation in which a robot assists a person with replacing a fuel injector on a ship using English, French, and Polish. In brief, the LEIA's ontological script for replacing fuel injectors — which gives it the knowledge it needs to reason about the scenario — goes as follows: HUMAN identifies and tags the intended fuel injector. LEIA shuts off the fuel valve. HUMAN

[9] For previous work on generation for LEIAs, see McShane & Leon (2021), McShane, Nirenburg, & English (2024, section 4.3), and McShane et al. (2025b).

checks that it's off and confirms that. LEIA removes the fuel line connection. LEIA removes the old fuel injector. HUMAN inspects it and confirms that. LEIA inserts the new fuel injector. HUMAN makes sure it is tight and confirms that. LEIA reconnects the fuel line. HUMAN tests it. LEIA leverages this knowledge to support reasoning in the following scenario. Tony, the human, is the speaker. LEIA backchannels by behaving appropriately.

1. "**We're going to replace a fuel injector**." [French: *Nous*[10] *allons remplacer un injecteur de carburant.* Polish: *Wymienimy wtryskiwacz paliwa*.] LEIA understands this as an indirect speech act (PROPOSE-PLAN), it infers who *we* is (speaker + me), and it creates a plan from its script that assigns agent roles appropriately. Before they start, Tony, the chief engineer, notices that another team in the engine room needs help.
2. "**But first, the girls need the short screwdriver**." [French: *Mais d'abord, les filles ont besoin du tournevis court*. Polish: *Ale najpierw dziewczyny potrzebują krótkiego śrubokręta*.] LEIA understands this as an indirect speech act (REQUEST-ACTION), it infers who is meant, and it postpones its other plan to carry this one out. By the time LEIA returns, Tony has identified and labeled the intended fuel injector.
3. "**Go ahead and start**." [French: *Vas-y, commence*. Polish: *No to zaczynaj*.] LEIA needs to resolve the ellipsis (start *what?*) based on the next step in the plan, which is turning off a valve. According to its plan, LEIA is the agent of that action and it carries it out. The next move is Tony's: he has ensure that the valve is fully off and let LEIA know that.
4. "**OK**." [French: *C'est bon*. Polish: *W porządku*.] In all our languages, this utterance could be used to express multiple speech acts, LEIA uses the expectations recorded in its plan to interpret it as giving the go-ahead. LEIA carries out several more physical actions. Then it's Tony's turn to do something.
5. "**We need to be sure the bore is clean**." [French: *Nous devons nous assurer que l'alésage est propre*. Polish: *Musimy się upewnić, że otwór jest czysty*.] This information isn't actually needed by the LEIA, since it knows to wait until Tony has given the go-ahead that the bore is clean; but people say all kinds of things that aren't strictly necessary and LEIA is prepared to understand their role in the dialog.
6. "**It's clean**." [French: *Il est propre*. Polish: *Jest czysty*.] This is the precondition for LEIA's next sequence of physical actions, some of which are interspersed with necessary checks and OKs by Tony. Tony carries out the final actions himself.
7. "**All set**." [French: *C'est tout*. Polish: *Gotowe*.] LEIA understands this as checking the plan as completed.

Fig. 2 provides select screen shots from the simulation, and Fig. 3 shows select under-the-hood panels from the simulation environment.

[10] The more colloquial pronoun *on* could be used instead of *nous*.

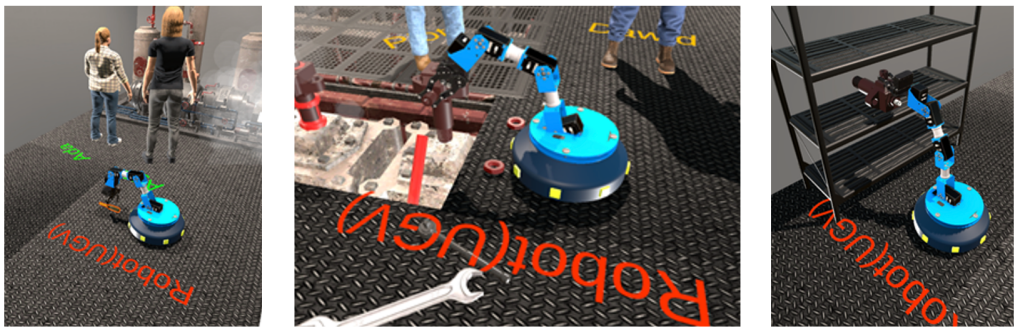


Fig. 2. Some screen shots from the shipboard simulation.

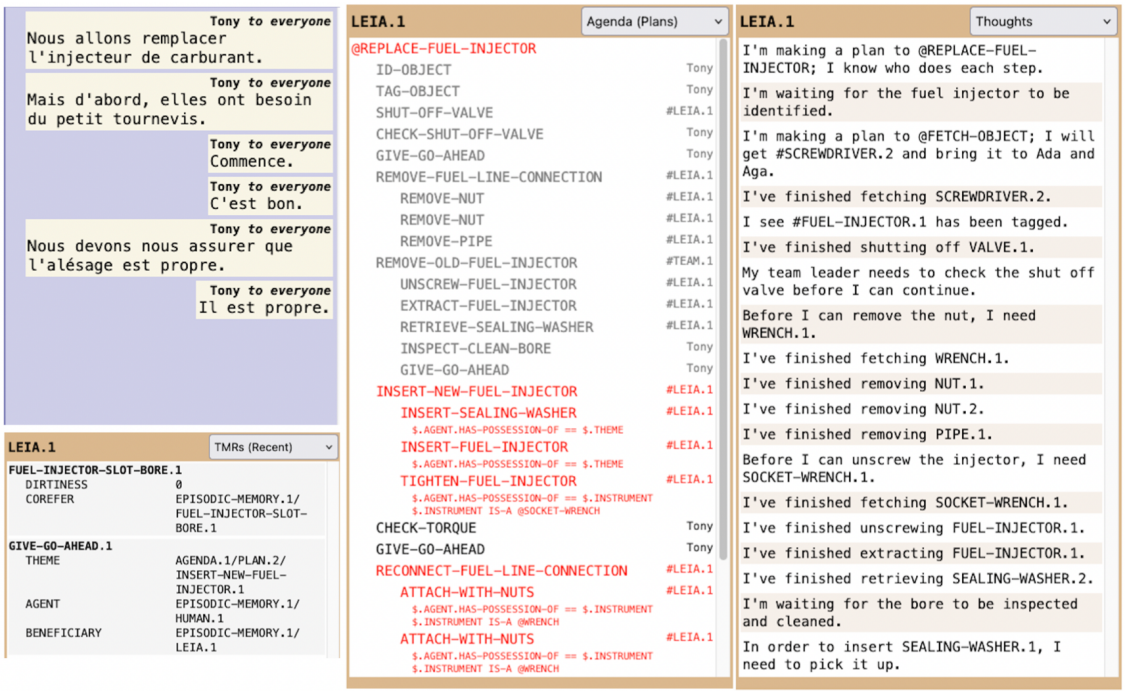


Fig. 3 Select under-the-hood panels from the French simulation run.

Under-the-hood panels show traces of system functioning rendered in a way that is understandable to people. Showing such traces is a key methodological tenet of LEIA development, as it offers transparency and explainability.

It is noteworthy that most of the challenges in this scenario involve language-independent reasoning, which is identical across language contexts and is what we expect to be the norm in real life. In other words, the challenging parts of operating in the world — for both humans and agents — involve mindreading, inferencing, planning, collaboration, and so on. Language processing, no matter the language, is a necessary support function that must not get in the way.

5 Comparisons with Others

We know of no other work that closely compares to what we have described. Distant resonances can be found in cross-linguistic semantic annotation, which prepares training sets for machine learning (Wein & Bonn, 2023), and in the evaluation of empirical MT for written texts (Ataman, et al., 2025). There has also been some recent work on multilingual robotics in empirical AI, for which three examples will suffice as illustration.

ReLI (short for "**Re**gardless of the **L**anguage of task **I**nstructions") is a system designed to allow a physically mobile robot to understand and carry out commands in any human language (Nwankwo et al., 2025). It is an LLM-based approach — one dependent on large-scale, pretrained LLMs — that has the expected limitations of reliance on this technology, such as the propensity for hallucinations, the opaqueness of reasoning, and the requirement of access to the cloud, which creates accessibility and latency issues in practical applications. The developers report "Across all benchmarked languages, ReLI achieved, on average, $90\% \pm 0.2$ accuracy in multilingual instruction parsing and task execution success rates. These results provide strong empirical evidence for ReLI and demonstrate the feasibility of cross-lingual language-to-action grounding across a broad range of linguistic conditions." Their interpretation of 90% accuracy as evidence of the feasibility of the approach is, in our opinion, not convincing since there is no compelling reason to believe that more use of the same methods will overcome the hardest 10% of challenges.

Chen et al. (2026) created a multilingual evaluation framework for Vision Language Action (VLA) models that were trained primarily on English. They used two simulation benchmarks that covered tasks such as object rearrangement and tool use. Their evaluation covered two settings: "The first setting is the multilingual instruction, where instructions are translated into different languages [using the Cloud Translation API] while preserving their original semantics, enabling controlled evaluation of multilingual generalization." Of course, as we have seen, machine translation does not reliably preserve the original semantics. "The second setting is the code-switching instruction, where instructions contain mixed-language expressions that commonly occur in multilingual communication scenarios." This setting relied on an LLM-based analysis procedure. The languages selected for evaluation were Chinese, French, Russian and Arabic. The overall assessment was that "models trained primarily on English instructions exhibit substantial performance degradation when evaluated on other languages." The authors propose ways to enhance performance; however, those methods, like the methods they seek to improve upon, are also empirical.

Finally, Dong et al. (2026) translated the LIBERO manipulation benchmark into ten languages and found that, in simulation settings, success rates dropped by 30-50%. They found varying rates of language sensitivity across tasks, leading them to suggest priorities for fine-tuning language-critical steps of procedures.

All three examples illustrate the familiar balance of benefits (broad coverage, immediate availability) and limitations (suboptimal accuracy, opaqueness) when using empirical methods for machine cognition.

6 Wrapping Up

If empirical AI could solve all problems of artificial intelligence, then the work of symbolic cognitive modeling would be strictly academic—not uninteresting, but pursued exclusively to satisfy human curiosity. However, empirical AI is demonstrably not enough if we consider reliability a central requirement — which, indeed, we must (e.g., Connell & Lynott, 2024; Cuskley, Woods, & Flaherty, 2024; Kapoor et al., 2024).

Our overarching claim is that human-inspired LEIA-style cognition offers a package of features — reliability, explainability, extensibility, and feasibility deriving from strong R&D methodology — that is well worth further exploration as a complement to large language models, whose costs and hard limitations (power usage, recalcitrant hallucinations) are not being sufficiently reckoned with. Modeling agents after humans has advantages that have been demonstrated here: the same knowledge bases and overall approach can be used for both language understanding and generation; reasoning about tasks and reasoning about language are integrated naturally; and a reasoning-centric approach to language processing allows agents to be multilingual in a way that mimics human polyglots—using the same knowledge core with different linguistic veneers.

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